Jim Pham
Quantitative Research
6/4/2007

Question #1

Compute the **mean**, **median,** and **mode** of the following performance evaluation scores.

Set of scores = X = {120, 110, 100, 115}

1.1 Mean

Formula: X' = ∑ X / N

Mean = X' = ∑ (120 + 110 + 100 + 115) / 4 = **111.25**

1.2 Median

The median is an average, defined as the score that divides a set of scores into two equal halves.

Sorted Distribution of Scores
100
110
115
120

Median = (110 + 115) / 2 = **112.5**

Note: A set of scores has an even number of observations; the median is the average of the two middle scores in the sorted list.

1.3 Mode

This set of scores has no mode because no score is most frequent.

Question #2

Compute the **variance** and **standard deviation** of those scores.

2.1 Variance

Formula: V= ∑(X- X')^2 / N

Number of scores = N = **4**
Mean = 111.25

X	(X-X')	(X-X')^2
100	-11.25	126.56
110	-1.25	1.56
115	3.75	14.06
120	8.75	76.56

Variance = V = ∑(126.56 + 1.56 + 14.06 + 76.56) / 4 = 218.74/4 = **54.69**

2.2 Standard Deviation

Formula : SD = sqrt(V)

Standard Deviation = SD = sqrt (54.69) = **7.39**

Question #3

The following scores are from the “Happy Home” test designed to measure the satisfaction of residents. (A high score indicates a high level of satisfaction in the services provided by the home.)

Set of scores = X = {50, 40, 30, 25, 15}

Compute the **mean**, **media** and **mode**.

3.1 Mean

<u>*Formula:*</u> X’ = $\sum$ X / N

Number of scores = N = **5**

Mean = X’ = $\sum$ (50 + 40 + 30 + 25 + 15) / 5 = 160/5 = **32**

3.2 Median

Sorted Distribution of Scores:
15
25
30 ← median
40
50

Median = **30**

3.3 Mode

This set of scores has no mode because there is no score that is most frequent.

3.4 Standard Deviation

Check the standard deviation using the rough-check formula for the SD: (H-L)/3
(The highest score in the set minus the lowest score)/3

SD = (H-L)/3 = (50-15) / 3 = **11.66**

3.5 Compute the variance and standard deviation for these scores.

3.5.1 Variance

Formula: V = $\sum(X- X')^2$ / N

Number of scores = N = **5**
Mean = X' = **32**

X	(X-X')	(X-X')^2
15	-17	289
25	-7	49
30	-2	4
40	8	64
50	18	324

Variance = V = $\sum$(324 +64 + 4 +49 +289) / 5 = 730/5 =**146**

3.5.2 Standard Deviation

Formula: SD = sqrt (V)

Variance = V = **146**
Standard deviation = SD = sqrt (146) = **12.08**

EXTRA CREDIT
Create a set of scores with a Standard Deviation of 2.0. Demonstrate that the set of scores has a standard deviation of 2.0 by computing that statistic. Show all steps of your work.

Hint: you can create a set with a small number of numbers, as long as it's balanced.

(I solved this problem by reversing the process.)

4.1 Standard Deviation

Standard deviation = SD = sqrt(V)
Variance = V = **4**
SD = sqrt(4) = **2**

4.2 Variance

The variance is the square of the standard deviation (i.e. SD^2 = 2*2 = 4). The number 4 can be derived from the following:

Number of scores = N = **7**
Mean = X' = **4**

Note: I can choose a set of 7 scores. Theoretically, you can choose any number that is divisible by 4 that will work (i.e. …16, 20, 24, 28…)

X	**(X-X')^2**	**(X-X')^2**
1	-3	9
2	-2	4
3	-1	1
4	0	0
5	1	1
6	2	4
7	3	9

<u>*Formula:*</u> V = ∑(X- X')^2 / N

Variance = V = ∑(9 +4 + 1 +0 +1+4+9) / 7 = 28/7 = **4**

4.3 Mean

<u>*Formula:*</u> X' = ∑ X / N

Set of scores = X = {1, 2, 3, 4, 5, 6, 7}
Mean = X' = ∑ (1 +2 +3 +4 +5 + 6 + 7) / 7
Mean = X' = ∑ 28/ 7 = **4**

4.4 Median (Odd number of scores)

Sorted Distribution of Scores
1
2
3
4 ← median
5
6
7

Medium = **4**

4.5 Mode

There is no score that is most frequent, and therefore, there is no mode.

4.6 Set of scores

Set of scores = X = {1, 2, 3, 4, 5, 6, 7}
Number of scores = N = **7**